ACTIVITY 05

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1. Geospatial Data Visualization

Dataset

City	Latitude	Longitude	Population (Millions)
Chennai	13.08	80.27	11
Bangalore	12.97	77.59	13
Hyderabad	17.38	78.48	10
Mumbai	19.07	72.87	20
Delhi	28.61	77.21	32
Pune	18.52	73.85	7
Kolkata	22.57	88.36	15
Ahmedabad	23.02	72.57	8
Jaipur	26.91	75.79	4
Kochi	9.93	76.26	2

Questions

Procedure

Import the required libraries and create the city dataset with latitude, longitude, and population values.

Use longitude and latitude values to plot city locations using a scatter/bubble map.

Represent population using bubble size and color, and add city labels using geom_text().

Analyze the visualization to identify the city with the highest population.

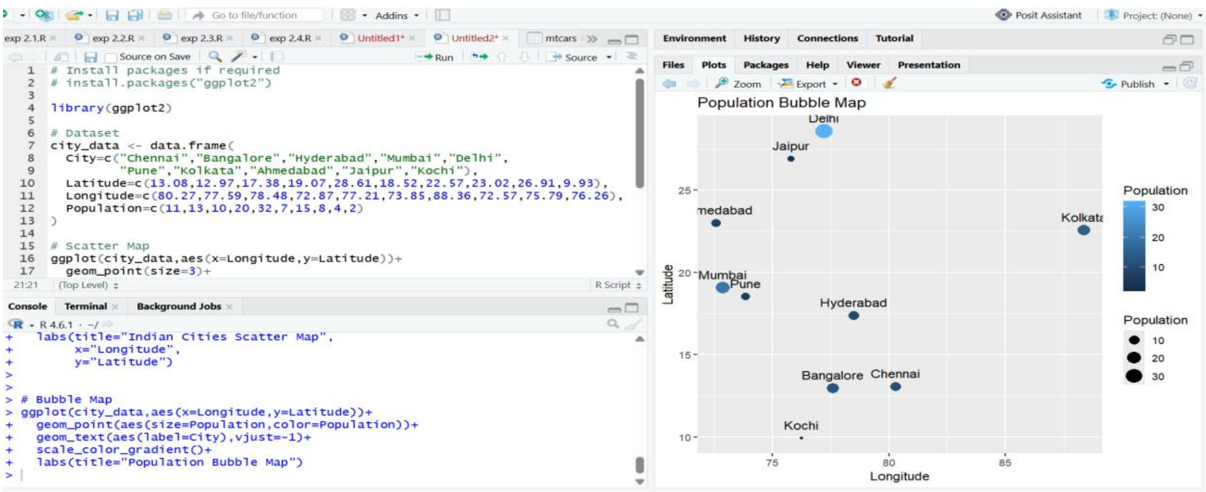
CODE:

```
# Install package if required

# install.packages("ggplot2")

library(ggplot2)

# Dataset
```

```
city_data <- data.frame(
  City=c("Chennai","Bangalore","Hyderabad","Mumbai","Delhi",
         "Pune","Kolkata","Ahmedabad","Jaipur","Kochi"),
  Latitude=c(13.08,12.97,17.38,19.07,28.61,
             18.52,22.57,23.02,26.91,9.93),
  Longitude=c(80.27,77.59,78.48,72.87,77.21,
              73.85,88.36,72.57,75.79,76.26),
  Population=c(11,13,10,20,32,7,15,8,4,2)
)
# Scatter + Bubble Map
ggplot(city_data,
      aes(x=Longitude,y=Latitude))+
  geom_point(aes(size=Population,
                 color=Population))+
  geom_text(aes(label=City),
            vjust=-1)+
  labs(title="Indian Cities Population Map",
      x="Longitude",
      y="Latitude")
```

Q1. Create a scatter map showing all cities using longitude and latitude.

Q2. Plot a bubble map where bubble size represents population.



Q3. Label each city on the plot using geom_text().

Answer: geom_text() is used to add labels to data points in ggplot2. It helps identify each city clearly on the map.

Q4. Use different colors based on population size.

Answer: Colors are used to represent different population levels. It helps compare cities based on population size.

Q5. Identify the city with the highest population from the visualization.

Answer: From the visualization, Delhi has the highest population with 32 million people.

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2. Network Visualization

Dataset

From	To
A	B
A	C
B	D
C	D
C	E
D	E
E	F
F	G
G	H
H	A

Questions

Procedure

Create the network dataset with source and destination nodes.

Use the igraph package to construct the network graph from edges.

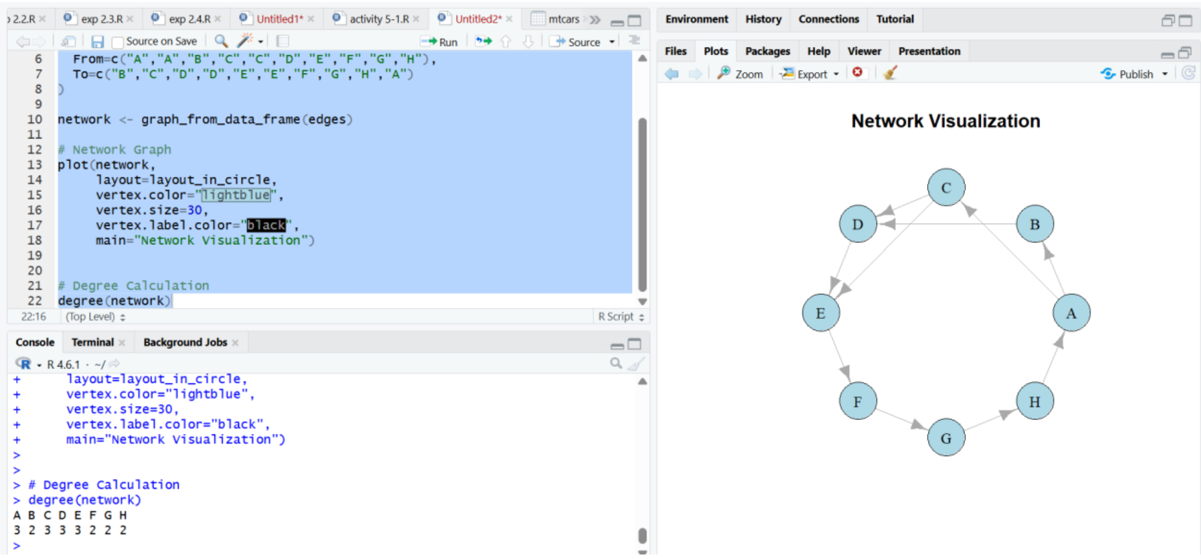
Customize node colors, labels, and display the graph using a circular layout.

Calculate node degree values and identify the most connected node.

CODE:

```
# Install package if required
# install.packages("igraph")
library(igraph)
# Dataset
edges <- data.frame(
  From=c("A","A","B","C","C","D","E","F","G","H"),
  To=c("B","C","D","D","E","E","F","G","H","A")
)
# Create Network
network <- graph_from_data_frame(edges)
# Network Graph
plot(network,
     layout=layout_in_circle,
     vertex.color="lightblue",
     vertex.size=30,
     vertex.label.color="black",
     main="Network Graph")
# Degree of nodes
degree(network)
```

Q1. Create a network graph using the igraph package.



Q2. Display node labels and customize node colors.

Answer: Node labels identify each node, while colors improve the readability of the network graph.

Q3. Calculate the degree of each node

Answer: Degree represents the number of connections linked to a node. It helps measure node importance in a network.

Q4. Identify the node with the highest number of connections.

Answer: Nodes A, C, D, and E have the highest number of connections.

Q5. Plot the network using a circular layout.

Answer: Circular layout arranges nodes in a circle. It provides a clear view of network relationships.

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3. Temporal Data Representation

Dataset

Month	Sales (₹ Lakhs)
Jan	120
Feb	135
Mar	150
Apr	170

Month	Sales (₹ Lakhs)
May	190
Jun	220
Jul	210
Aug	230
Sep	250
Oct	270
Nov	290
Dec	320

Questions

Procedure

Create the monthly sales dataset with month names and sales values.

Plot sales data using a line chart to show changes over time.

Customize the graph using axis labels, data points, and grid lines.

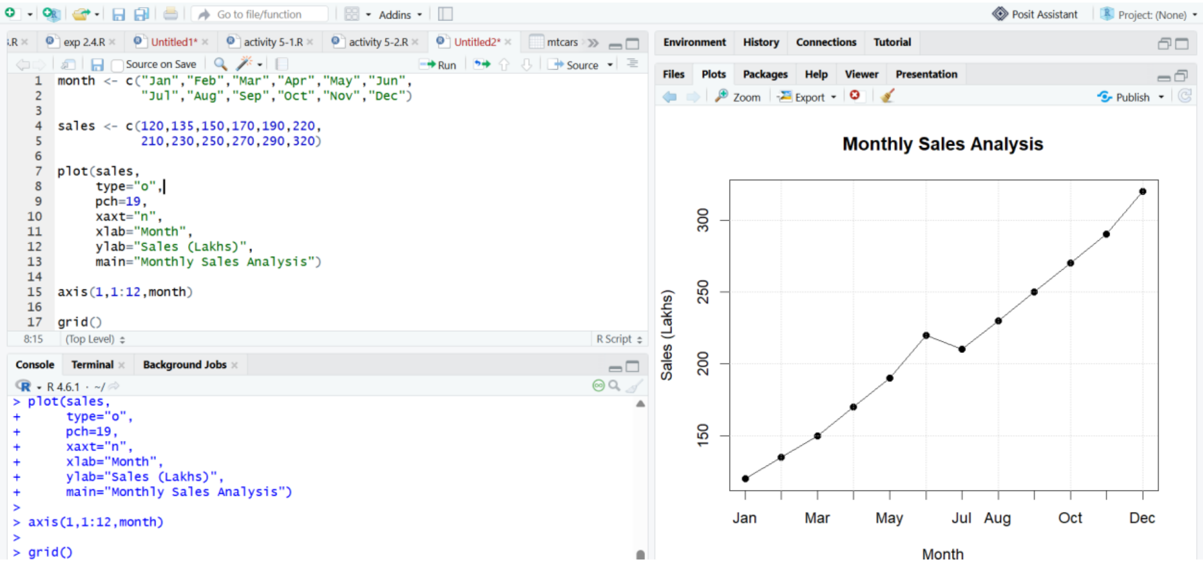
Analyze the graph to find the month with maximum sales.

CODE

```
# Dataset
month <- c("Jan","Feb","Mar","Apr",
           "May","Jun","Jul","Aug",
           "Sep","Oct","Nov","Dec")
sales <- c(120,135,150,170,190,220,
           210,230,250,270,290,320)
# Line Chart
plot(sales,
     type="o",
     pch=19,
     xaxt="n",
```

```
    xlab="Month",
    ylab="Sales (Lakhs)",
    main="Monthly Sales")
# Add month labels
axis(1,
    at=1:12,
    labels=month)
# Grid lines
grid()
```

Q1. Create a line chart showing monthly sales.



Q2. Add custom month names using the axis() function.

Answer: The axis() function is used to customize axis labels. It replaces default values with month names.

Q3. Highlight data points using pch=19.

Answer: pch=19 creates filled circular points in R plots. It highlights individual data values.

Q4. Display grid lines and proper axis labels.

Answer: Grid lines improve graph readability. Axis labels provide information about the displayed variables.

Q5. Identify the month with the maximum sales from the graph.

Answer: December has the maximum sales of ₹320 Lakhs.

4. Multivariate Data Visualization

Dataset

Student	Marks	Attendance (%)	Study Hours	Projects
A	60	75	2	1
B	68	80	3	2
C	74	84	4	2
D	79	88	5	3
E	82	90	5	3
F	85	92	6	4
G	88	94	6	4
H	91	95	7	5
I	94	97	8	5
J	97	99	9	6

Questions

Procedure

Create the student dataset containing marks, attendance, study hours, and projects.

Generate a bubble plot and scatter plot matrix to visualize relationships among variables.

Calculate correlations and represent relationships using a heatmap.

Analyze the graphs to identify patterns and the student with highest performance.

CODE:

```
# Dataset

student <- data.frame(

 Student=c("A","B","C","D","E",

           "F","G","H","I","J"),

 Marks=c(60,68,74,79,82,

         85,88,91,94,97),

 Attendance=c(75,80,84,88,90,
```

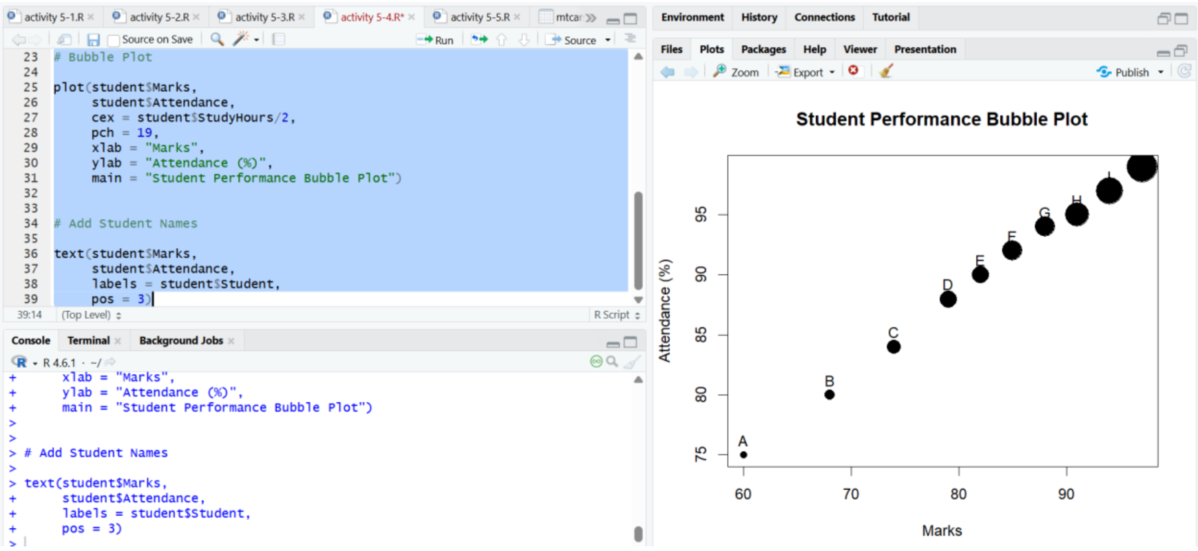
```
             92,94,95,97,99),
 StudyHours=c(2,3,4,5,5,
             6,6,7,8,9),
 Projects=c(1,2,2,3,3,
           4,4,5,5,6)
)
# Q1 Bubble Plot
plot(student$Marks,
     student$Attendance,
     cex=student$StudyHours/2,
     pch=19,
     xlab="Marks",
     ylab="Attendance (%)",
     main="Student Bubble Plot")
text(student$Marks,
     student$Attendance,
     labels=student$Student,
     pos=3)
# Q2 Scatter Plot Matrix
pairs(student[,2:5],
      main="Student Performance Matrix")
# Q3 Correlation
cor(student[,c("Marks",
               "Attendance",
               "StudyHours")])
# Q4 Highest Marks Student
student[which.max(student$Marks),]
# Q5 Heatmap
heatmap(cor(student[,2:5]),
```

main="Correlation Heatmap",

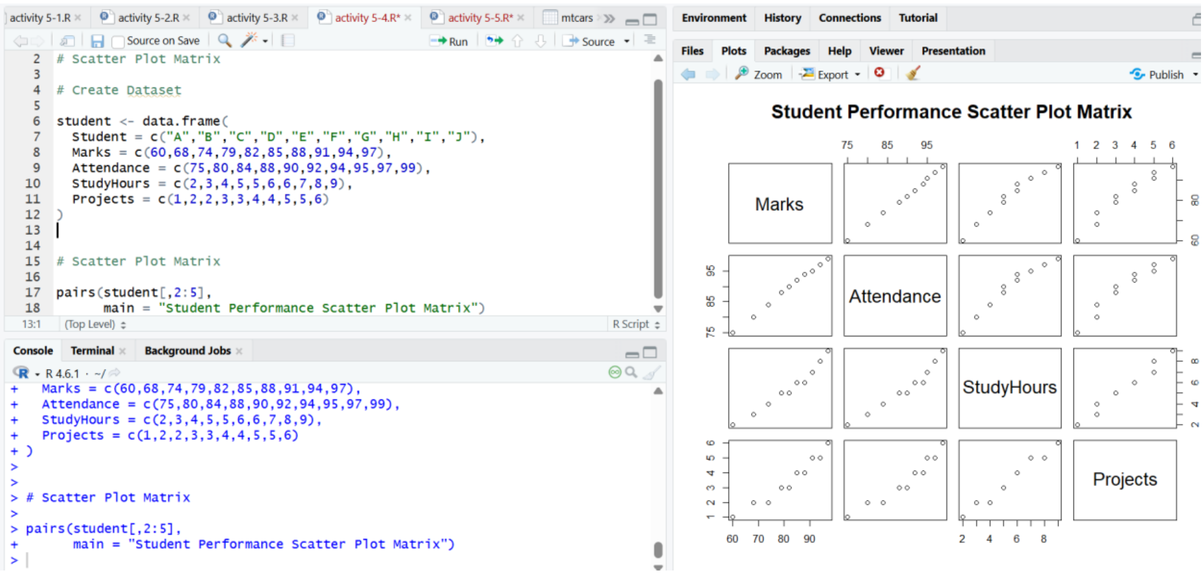
scale="none")

Q1. Create a bubble plot where:

- **X-axis = Marks**
- **Y-axis = Attendance**
- **Bubble size = Study Hours**



Q2. Produce a scatter plot matrix using the pairs() function.



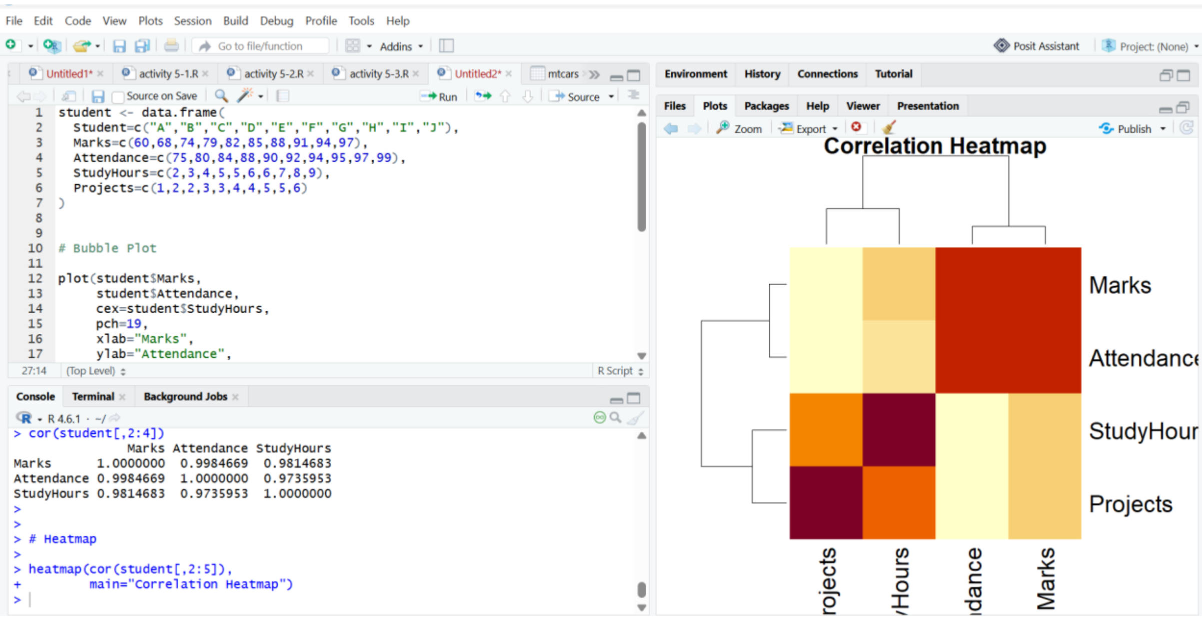
Q3. Find the correlation among Marks, Attendance, and Study Hours.

Answer: Correlation measures the relationship between variables. Positive correlation shows variables increase together.

Q4. Identify the student with the highest marks and largest bubble.

Answer: Student J has the highest marks (97) and largest bubble size.

Q5. Create a heatmap showing the relationships among all numerical variables.



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5. Integrated Data Visualization Exercise

Dataset

Year	Sales	Profit	Customers	Branches
2016	150	20	1200	5
2017	170	25	1500	6
2018	200	32	1800	7
2019	240	40	2200	8
2020	260	42	2400	8
2021	300	55	2800	10
2022	340	65	3300	11
2023	390	72	3800	12
2024	430	82	4200	13
2025	480	95	4700	15

Questions

Procedure

Create the business dataset containing year, sales, profit, customers, and branches.

Visualize yearly sales using a line chart and create a bubble plot for sales-profit analysis.

Generate correlation matrix and scatter plot matrix to study variable relationships.

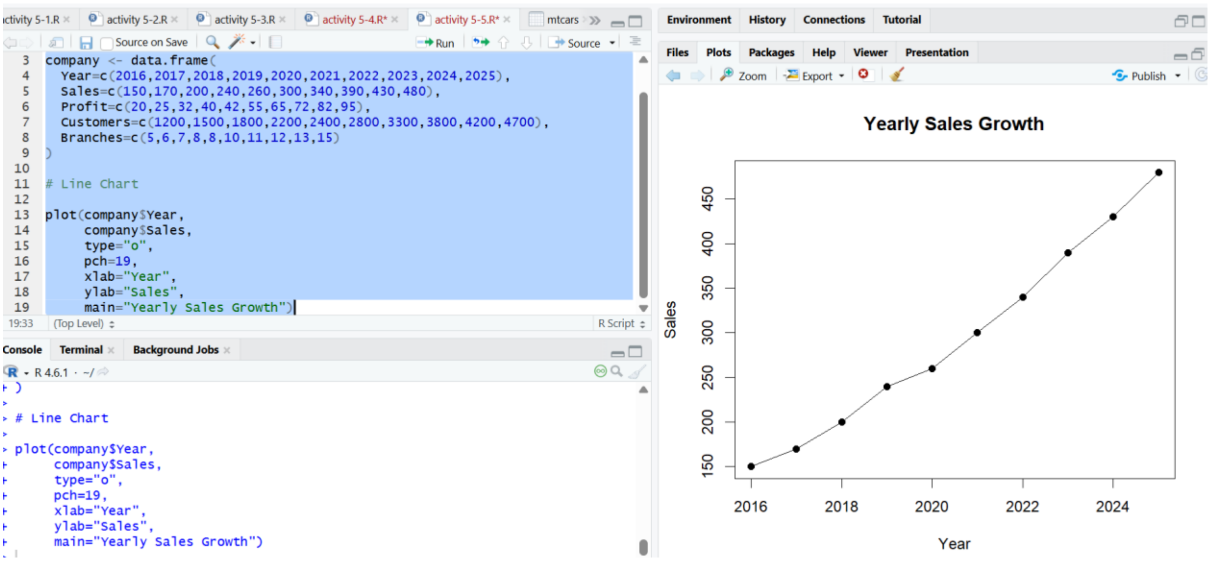
Interpret the visualizations to determine the relationship between branches, sales, and profit.

CODE

```
# Dataset
company <- data.frame(
Year=c(2016,2017,2018,2019,2020,
       2021,2022,2023,2024,2025),
Sales=c(150,170,200,240,260,
        300,340,390,430,480),
Profit=c(20,25,32,40,42,
         55,65,72,82,95),
Customers=c(1200,1500,1800,2200,2400,
            2800,3300,3800,4200,4700),
Branches=c(5,6,7,8,8,
           10,11,12,13,15)
)
# Q1 Line Chart
plot(company$Year,
     company$Sales,
     type="o",
     pch=19,
     xlab="Year",
     ylab="Sales",
     main="Yearly Sales Growth")
# Q2 Bubble Plot
plot(company$Sales,
```

```
    company$Profit,
    cex=company$Customers/1000,
    pch=19,
    xlab="Sales",
    ylab="Profit",
    main="Sales vs Profit Bubble Plot")
text(company$Sales,
    company$Profit,
    labels=company$Year,
    pos=3)
# Q3 Correlation Matrix
cor(company[,2:5])
# Q4 Scatter Plot Matrix
pairs(company[,2:5],
     main="Company Data Scatter Plot Matrix")
```

Q1. Create a temporal line chart showing yearly sales.



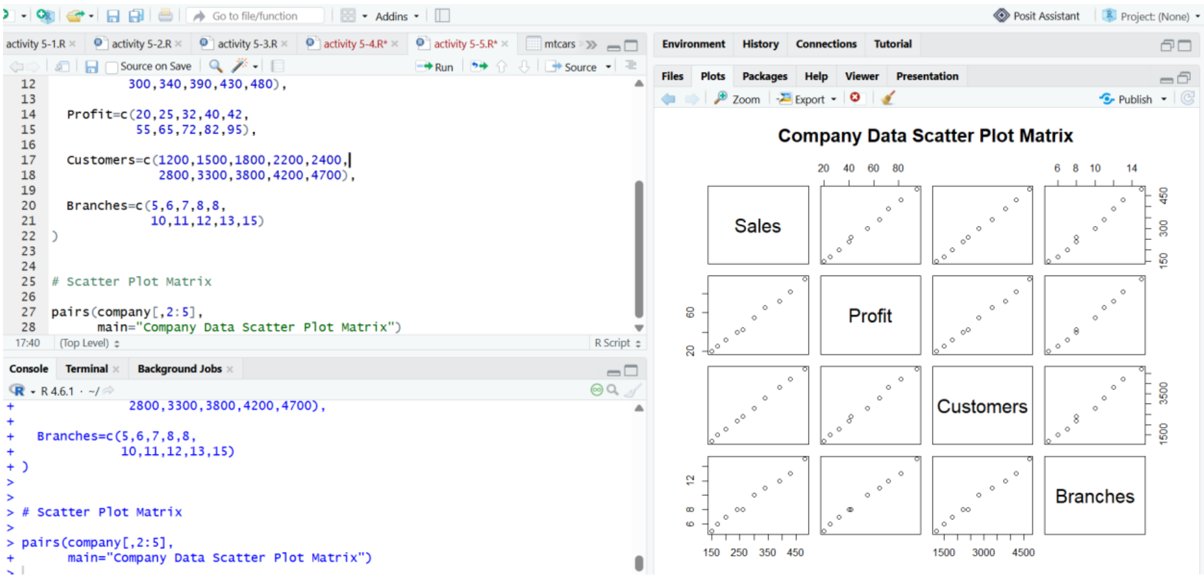
Q2. Generate a multivariate bubble plot with:

- **X-axis = Sales**
- **Y-axis = Profit**
- **Bubble size = Customers**

Q3. Compute the correlation matrix for Sales, Profit, Customers, and Branches.

Answer: Correlation matrix shows relationships among numerical variables. It helps identify positive or negative associations.

Q4. Create a scatter plot matrix using the pairs() function.



Q5. Interpret the visualizations to determine whether an increase in branches is associated

Answer: The graph shows that increasing branches leads to higher sales and profits. It indicates a positive relationship.with higher sales and profits.